

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Object Oriented Analysis and Design

COURSE CODE: CSA1105

COURSE OUTCOME COVERED

CO5: Apply object-oriented analysis and design principles along with appropriate software development life cycle models to real-world application scenarios. (BL3, BL4)

QUESTIONS: 5

Annexure A

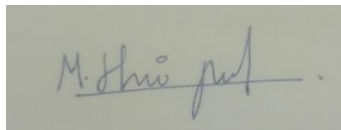
Constraint-Based Problem-Solving

Code of Conduct:

I Hariprasath(192421246) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'M. Hariprasath' followed by a horizontal line.

Course Faculty:

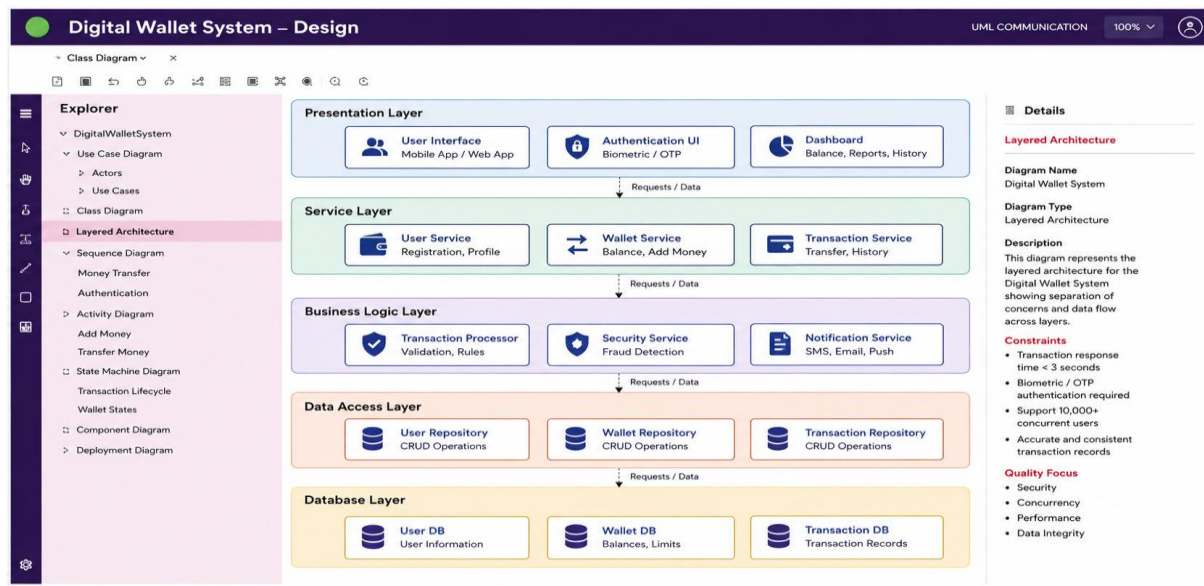
Dr. Ramamoorthy

ANNEXURE A

CONSTRAINT-BASED PROBLEM SOLVING

Q1. Digital Wallet System

Scenario: A digital wallet provides User Registration, Wallet Management, Money Transfer, and Transaction History. The system must respond to transactions within 3 seconds, require biometric or OTP authentication, support more than 10,000 concurrent users, and maintain accurate and consistent transaction records.



i. Quality Assurance Analysis: Quality assurance for the digital wallet focuses on functional correctness, security, performance, and data integrity. Testing should be performed throughout the development lifecycle so that defects are identified before they affect real transactions.

ii. Potential Defects: Security defects: OTP bypass, weak biometric validation, session hijacking, unauthorized wallet access, insecure APIs, and exposure of sensitive user or payment information. **Concurrency defects:** race conditions, double spending, two transfers using the same balance, and inconsistent balances when many users transact simultaneously.

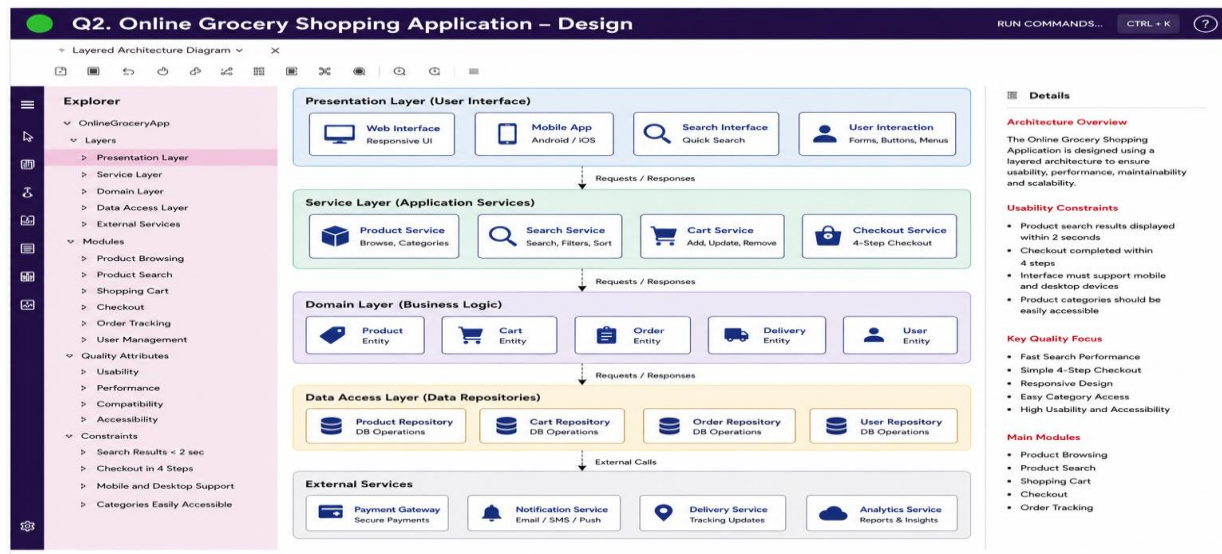
iii. Suitable Testing Strategies: Functional testing: Verify registration, authentication, adding money, transferring money, balance checking, and transaction-history operations.

iv. Justification

These techniques ensure reliability by checking the system from functional, security, performance, and data perspectives. Load and stress testing validate scalability, security testing protects user funds, concurrency testing prevents double spending, and transaction/database testing preserves accurate records.

Q2. Online Grocery Shopping Application

Scenario: The application provides Product Browsing, Product Search, Shopping Cart, Checkout, and Order Tracking. Search results must appear within 2 seconds, checkout must be completed within 4 steps, the interface must work on mobile and desktop, and product categories must be easy to access.



i. Usability Analysis: Usability testing evaluates how easily users can find products, add items to a cart, complete checkout, and track orders. The evaluation should measure task completion time, error rate, navigation difficulty, user satisfaction, and accessibility.

ii. Potential Usability Problems: Slow or unclear search results can make users wait or repeat searches. Too many checkout screens or unnecessary fields can increase abandonment. Poor responsive design may cause buttons, menus, or product information to be difficult to use on mobile devices.

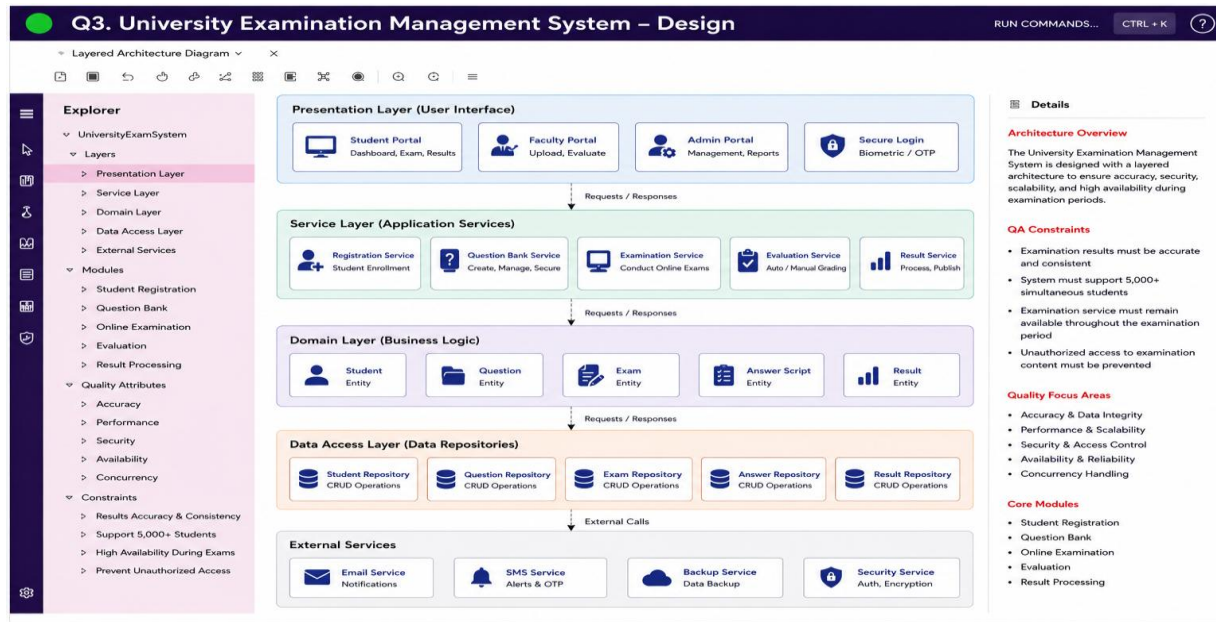
iii. Suitable Usability Testing Techniques: Ask representative users to search for a product, add it to the cart, complete checkout, and track an order. Ask users to describe what they expect while navigating the application to identify confusing controls and terminology.

iv. Recommended Improvements: Provide a prominent search bar with autocomplete, filters, sorting, and relevant suggestions. Display major product categories directly on the home page using clear labels. Use a short four-step checkout: Cart → Delivery Details → Payment → Confirmation.

v. Justification: The improvements reduce navigation effort and user errors while making important information easier to find. Fast search improves task efficiency, a four-step checkout reduces abandonment, responsive design improves mobile and desktop usability, and accessible controls allow more users to complete tasks independently.

Q3. University Examination Management System

Scenario: The system contains Student Registration, Question Bank, Online Examination, Evaluation, and Result Processing. Results must be accurate and consistent, more than 5,000 students must be supported simultaneously, the service must remain available during examinations, and unauthorized access to examination content must be prevented.



i. Quality Assurance Analysis

Because examinations are time-critical, QA must concentrate on accuracy, security, availability, performance, concurrency, and recovery. Defects in question delivery, answer submission, evaluation, or result processing can directly affect students

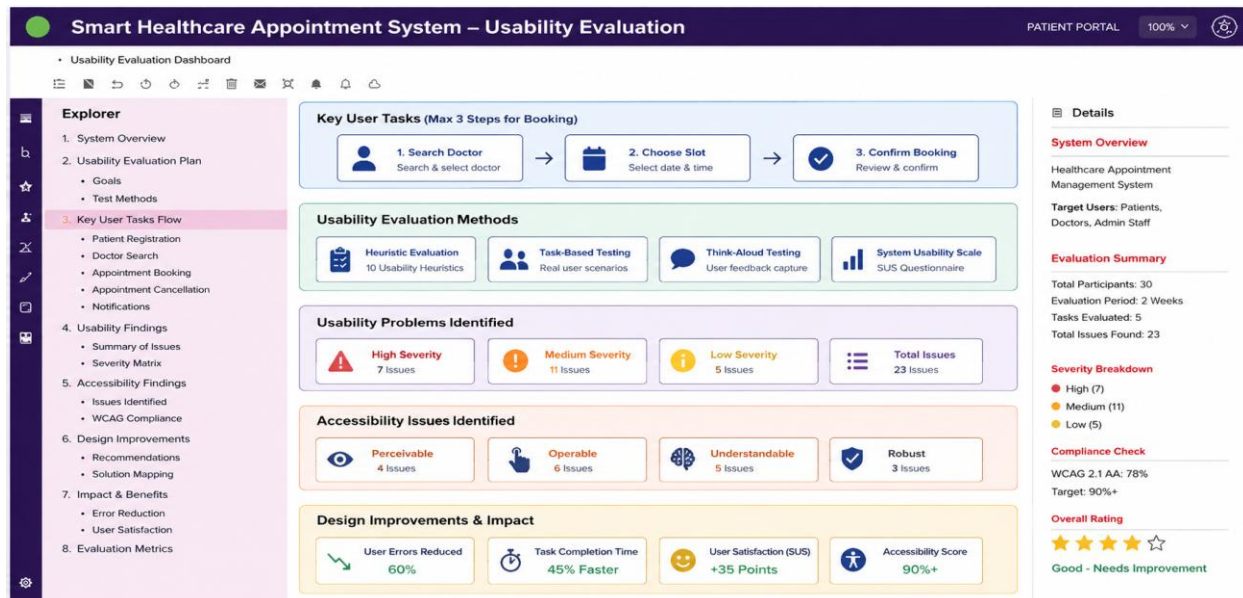
ii. Critical Test Scenarios: Verify that submitted answers are stored correctly, marks are calculated according to the answer key, negative marking rules are applied correctly when applicable, and the evaluation data.

iii. Suitable Testing Techniques: Functional and system testing: Verify every examination workflow from registration through result publication. Load and stress testing: Validate support for 5,000+ simultaneous students and identify capacity limits.

iv. Justification: These techniques reduce the risk of incorrect marks, examination disruption, unauthorized access, and data loss. Performance and concurrency testing confirm that thousands of students can use the system at the same time. Security testing protects examination content, while accuracy and database testing ensure that results are correct and consistent.

Q4. Smart Healthcare Appointment System

Scenario: The system provides Patient Registration, Doctor Search, Appointment Booking, Appointment Cancellation, and Notifications. Booking should require a maximum of 3 major steps, the interface should support users with different technical experience and confirmation messages should be easy to understand.



i. Usability Evaluation: Usability evaluation should determine whether patients can find a suitable doctor, understand available appointment slots, book or cancel an appointment, and understand confirmation messages without unnecessary assistance.

ii. Potential Usability and Accessibility Problems: Long registration or booking forms can increase abandonment and user errors. Unclear appointment availability may cause users to select unavailable or confusing time slots. Complex medical terminology or technical interface elements can confuse less experienced users.

iii. Suitable Testing Methods: Task-based usability testing: Ask users to search for a doctor, choose a slot, book an appointment, and cancel it. User observation and think-aloud testing: Observe where users hesitate, make mistakes, or misunderstand instructions.

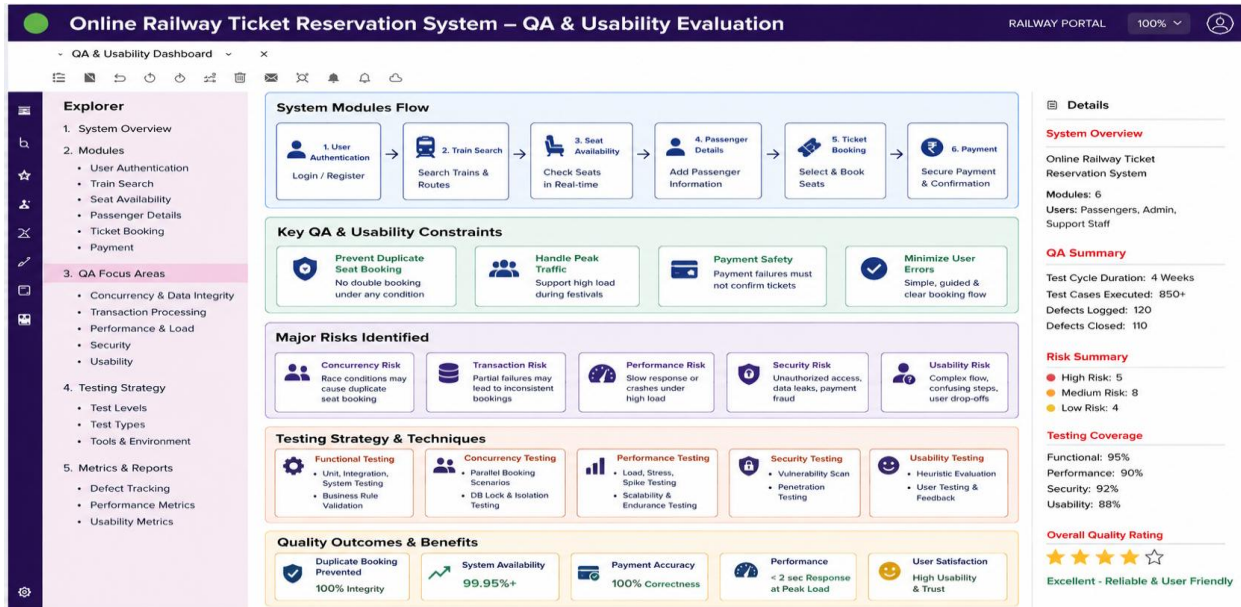
iv. Recommended Design Improvements: Use a simple three-step booking flow: Select Doctor → Select Available Slot → Confirm Appointment. Display available, unavailable, and selected slots clearly with text labels and accessible indicators rather than relying only on color.

v. Justification

A three-step flow reduces cognitive load and the chance of incorrect bookings. Clear availability information prevents selection errors, while plain language and accessible controls help users with different technical abilities.

Q5. Online Railway Ticket Reservation System

Scenario: The railway system includes User Authentication, Train Search, Seat Availability, Passenger Details, Ticket Booking, and Payment. Duplicate seat booking must be prevented, the system must handle festival season traffic, payment failures must not create incorrect confirmations, and the booking interface should minimize user errors.



i. QA and Usability Analysis

The railway system requires strong quality assurance because a booking may involve simultaneous users, limited seats, payment processing, and sensitive personal information. Testing must cover concurrency, transaction processing, performance, security, and usability.

ii. Major Risks: Concurrency risks: Two or more users may attempt to reserve the same seat at the same time, resulting in duplicate booking. Transaction risks: A payment may succeed while ticket creation fails, or ticket confirmation may appear before payment is completed.

iii. Suitable Testing Strategies: Concurrency testing: Simulate multiple users selecting and paying for the same seat and verify that only one valid booking is confirmed. Recovery testing: Interrupt payment or booking services and verify that seats are released or transactions are safely recovered without incorrect confirmation.

iv. Recommended Usability Improvements: Use a clear booking flow: Search Train → Select Seat/Class → Enter Passenger Details → Payment → Confirmation.

v. Justification: Concurrency and transaction testing prevent duplicate seats and incorrect ticket confirmations. Load and stress testing prepare the system for peak festival traffic. Security testing protects accounts and payment information.